

Upcoming Dates

- HW6 and Module 2 Post lab worksheet due today
- Module 3 Worksheet due Tuesday
 - Bring a copy to class for reference
- Proficiency testing – Tuesday
- Module 4 Prelab – Due Thursday, Feb, 26.
 - Begins March 2, signups next week.

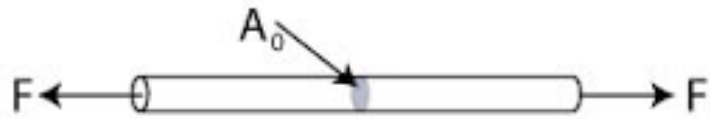
Exam 1

- **Thursday, March 5**
 - **Lectures 1-8**
 - **Lab Modules 2 and 3**
- **Includes proficiency attempt #2**

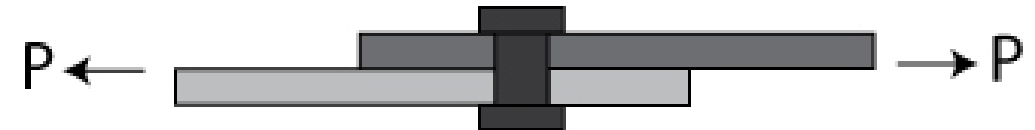
Strength of Materials:

Alternative Types of Stress

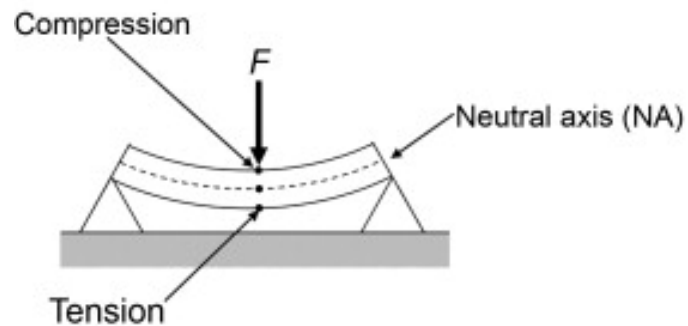
Axial Stress


$$\text{Stress, } \sigma = \frac{\text{Force}}{\text{Cross-Sectional Area}} = \frac{F}{A_0}$$

Simple Shear Stress


$$\tau = \frac{P}{A}$$

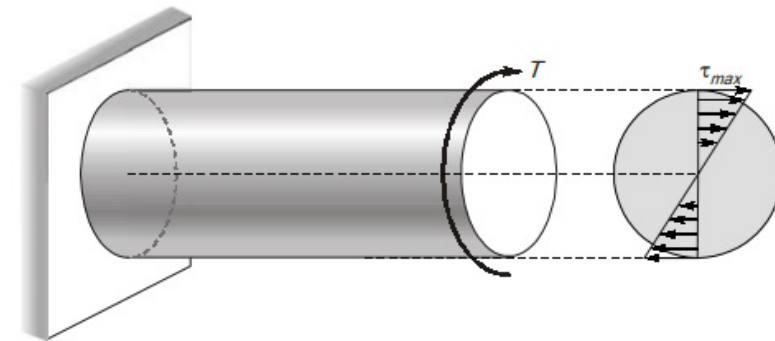
Bending Stress



Simple supported beam bending

$$\sigma = \frac{Mc}{I}$$

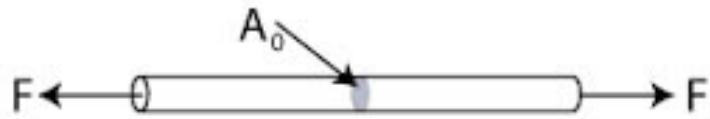
Torsional Shear Stress



$$\tau = \frac{Tr}{J}$$

Tensile Test

Axial Stress



The diagram shows a horizontal cylindrical specimen under tensile load. Two arrows labeled 'F' point outwards from the ends of the cylinder. A cross-section is highlighted in the middle, with a label 'A₀' and an arrow pointing to it.

$$\text{Stress, } \sigma = \frac{\text{Force}}{\text{Cross-Sectional Area}} = \frac{F}{A_0}$$

The tensile test → Only Axial Stress

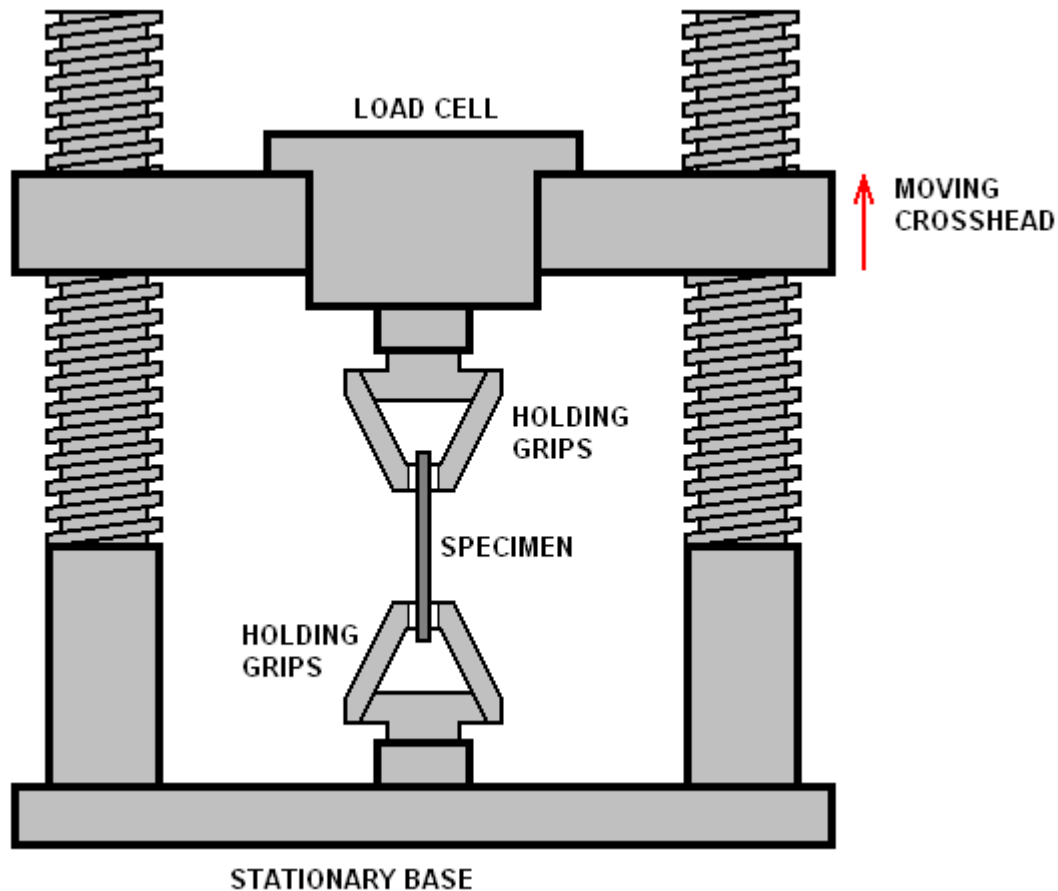
We use the properties found from Axial loading to evaluate other loading modes

Axial Stress is the same across the volume if there is a constant cross-section

For other types of loading, stress **varies by location**.

Stress-Strain for Ceramics

Metals – tensile test



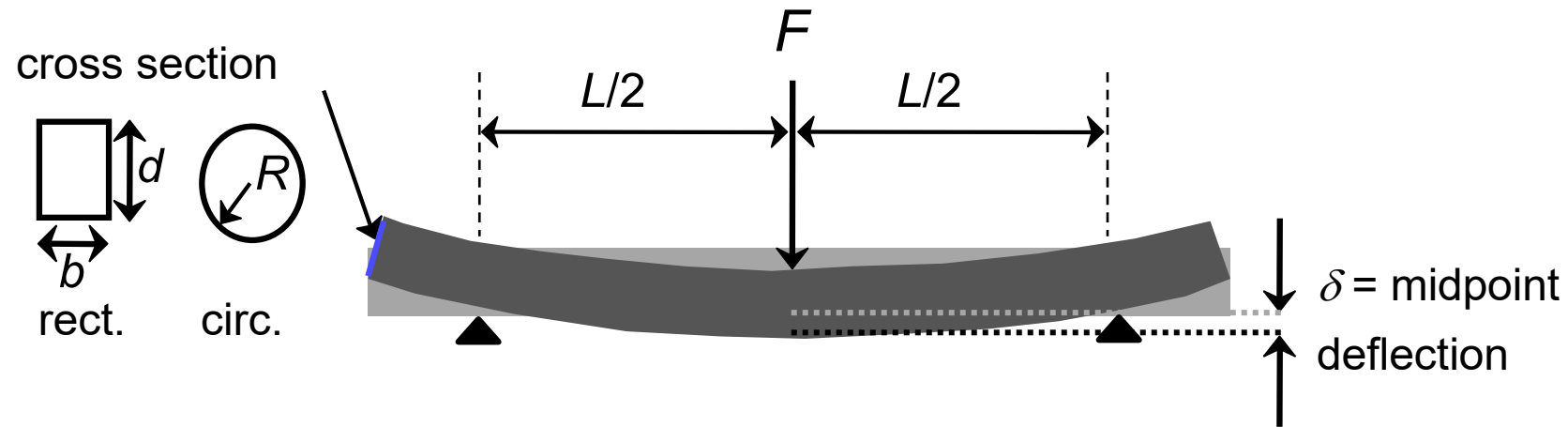
Ceramics – tensile test? → **NO!**

- Difficult to prepare the test sample with required geometry
- Difficult to grip brittle materials
- Grips must be perfectly aligned

Measurement of Elastic Modulus For Ceramics

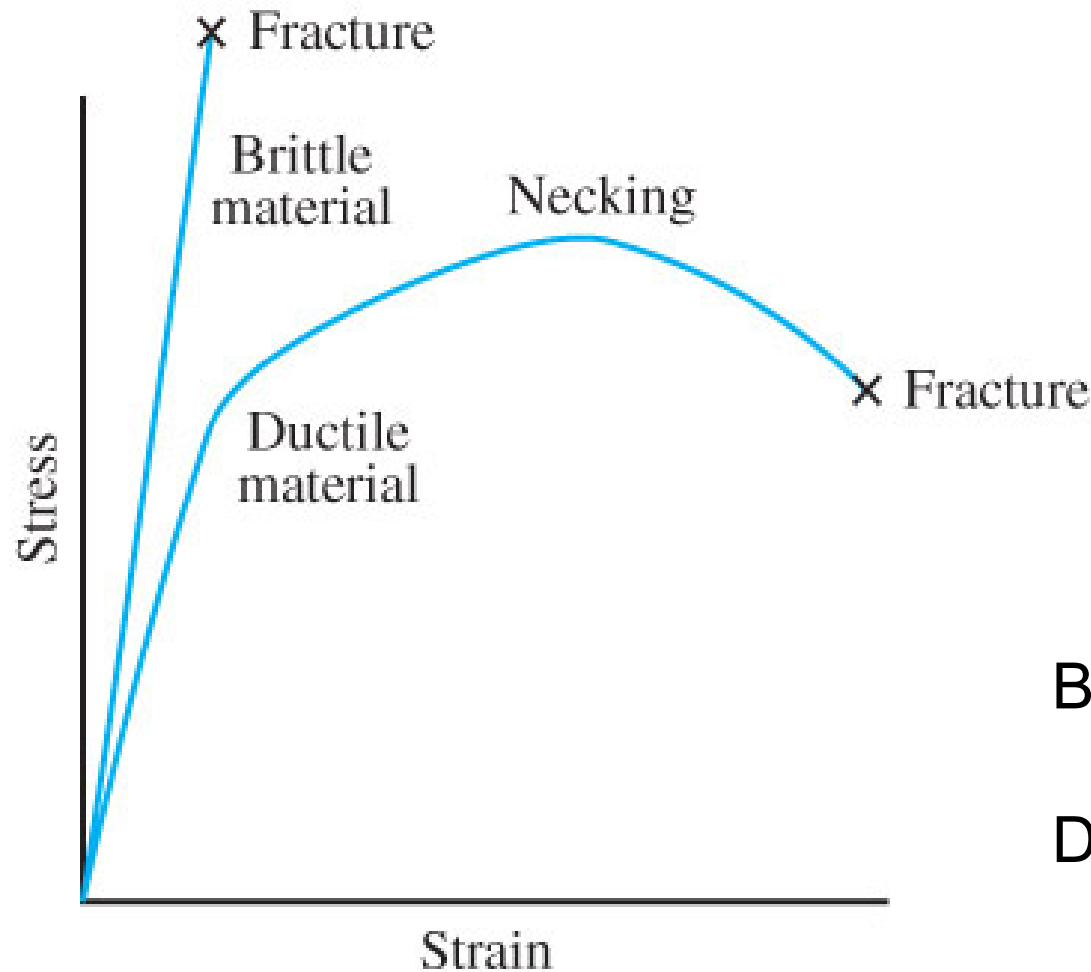
At room T , behavior is usually elastic, with brittle failure.

3-Point Bend Testing often used:



Young's Modulus, E , and Fracture stress can be obtained

Stress-Strain Elastic Behavior of Brittle Materials (Ceramics and Glasses)

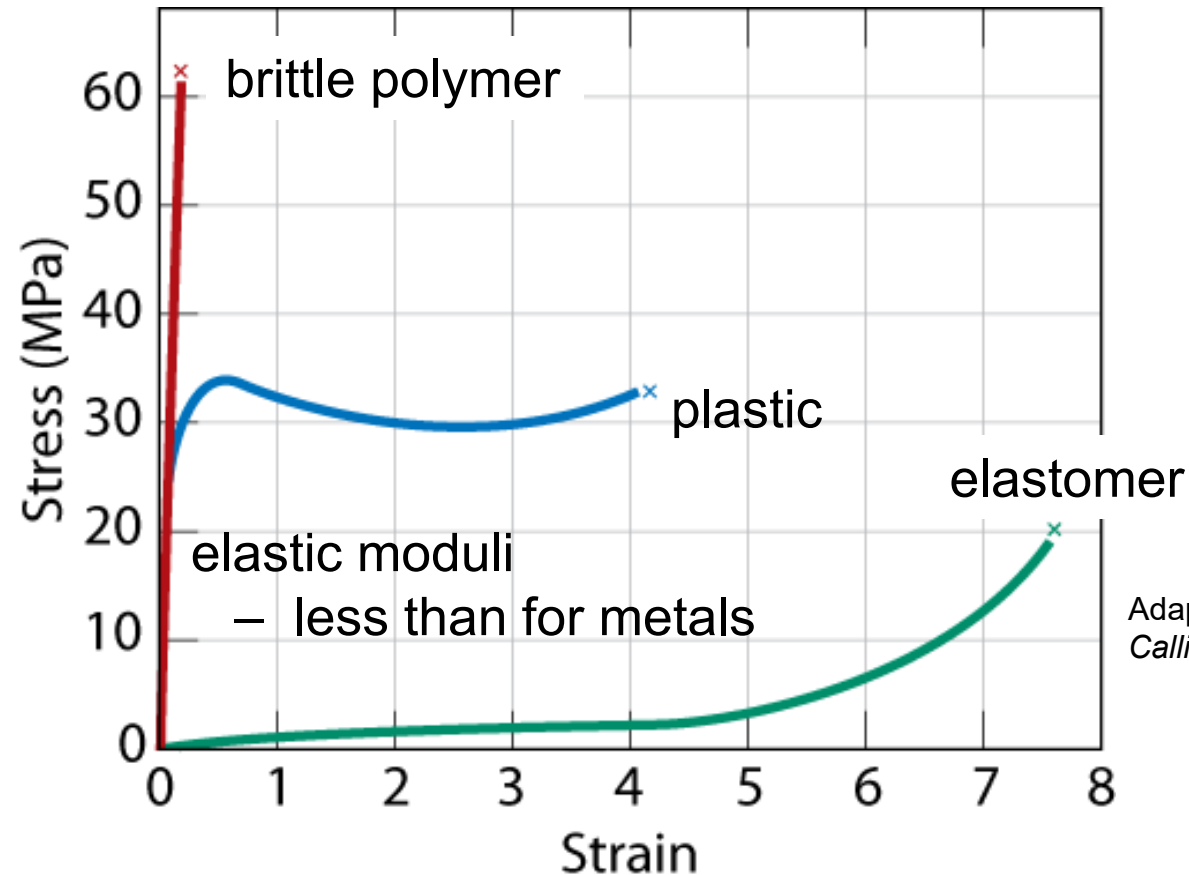


Brittle Materials: $< 5\%$ elongation

Ductile Materials: $\geq 5\%$ elongation

Figure 7.8

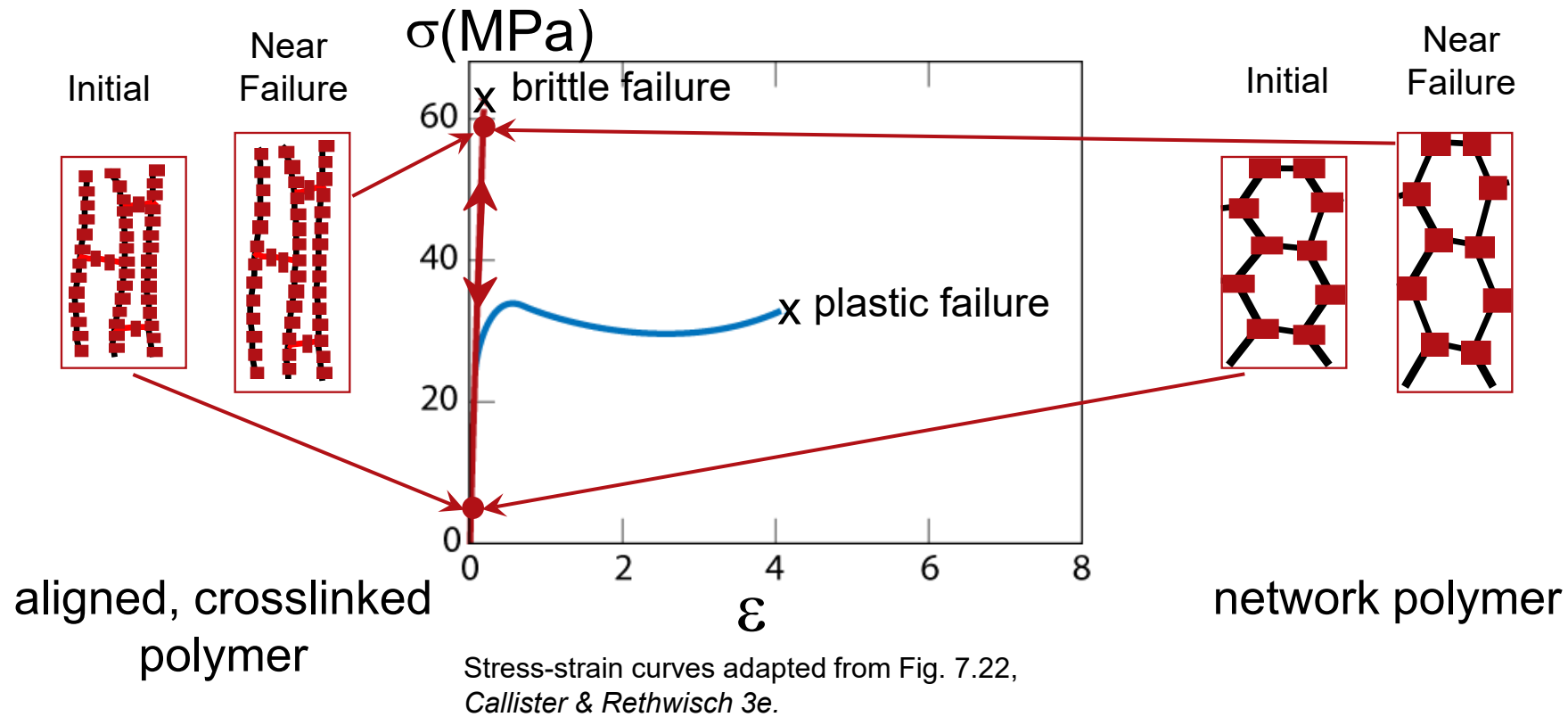
Mechanical Properties of Polymers – Stress-Strain Behavior



Adapted from Fig. 7.22,
Callister & Rethwisch 3e.

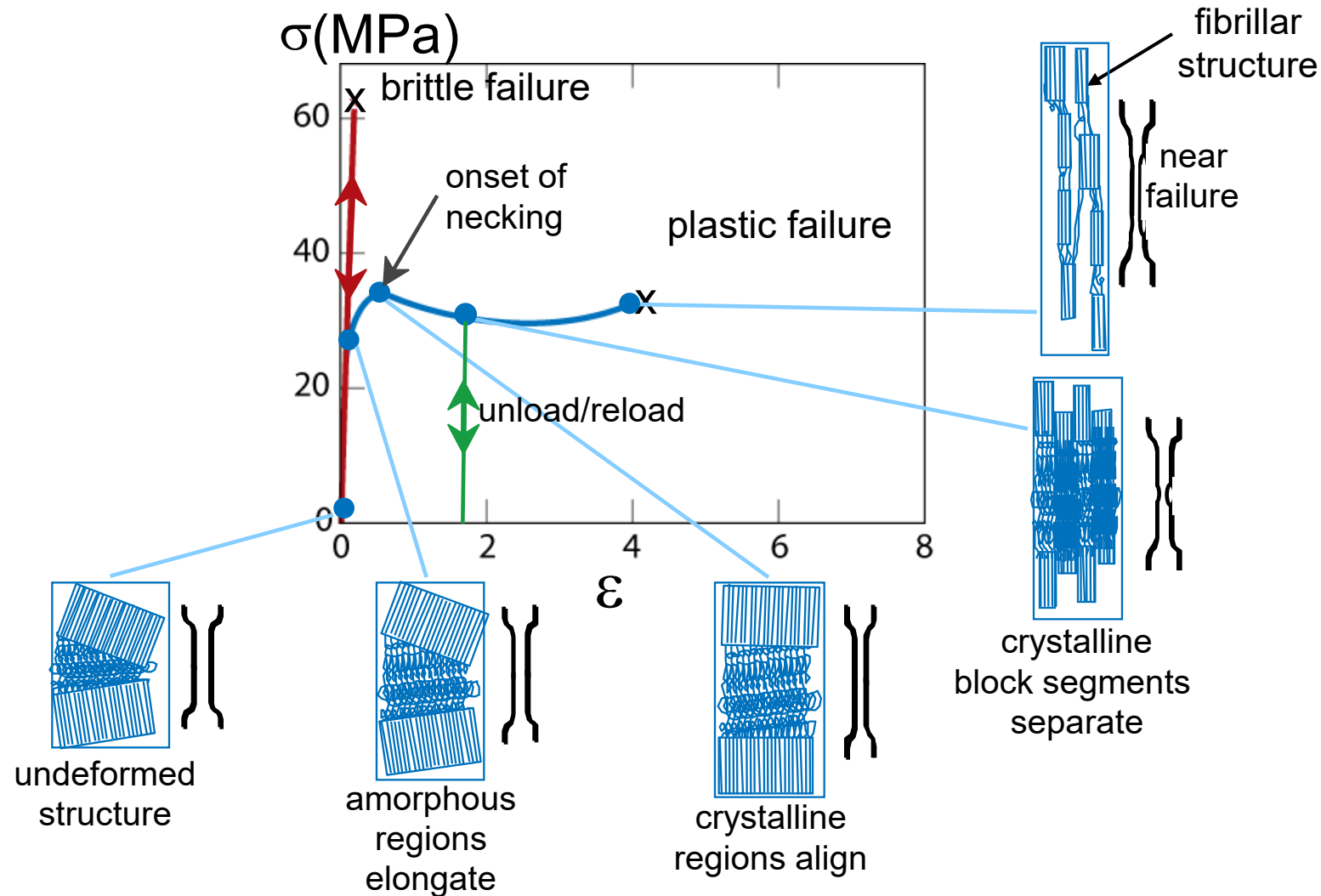
- Fracture strengths of polymers ~ 10% of those for metals
- Deformation strains for polymers > 1000%
 - for most metals, deformation strains < 10%

Mechanisms of Deformation—Brittle Crosslinked and Network Polymers



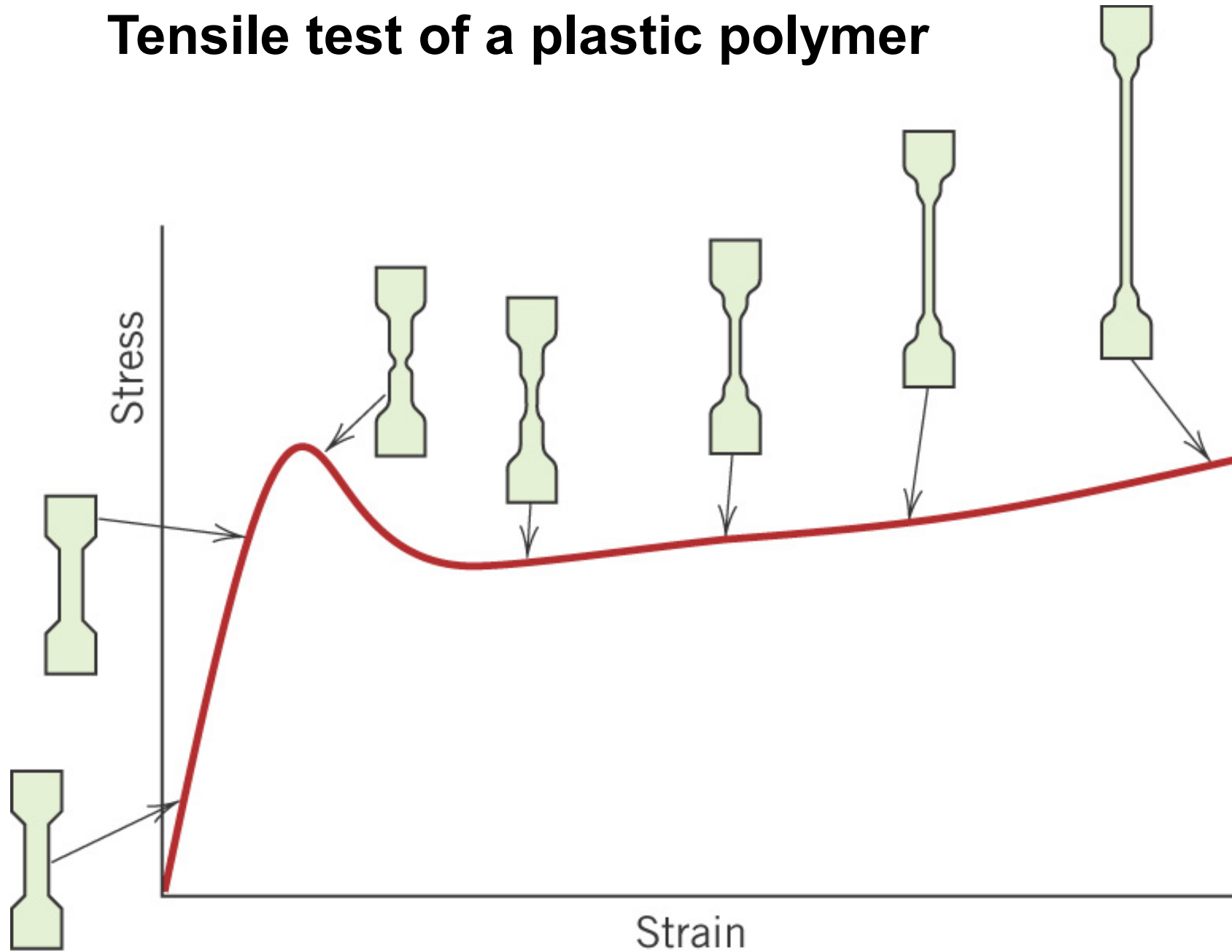
-- brittle behavior (of aligned, crosslinked & network polymers)

Mechanisms of Deformation — Semicrystalline (Plastic) Polymers

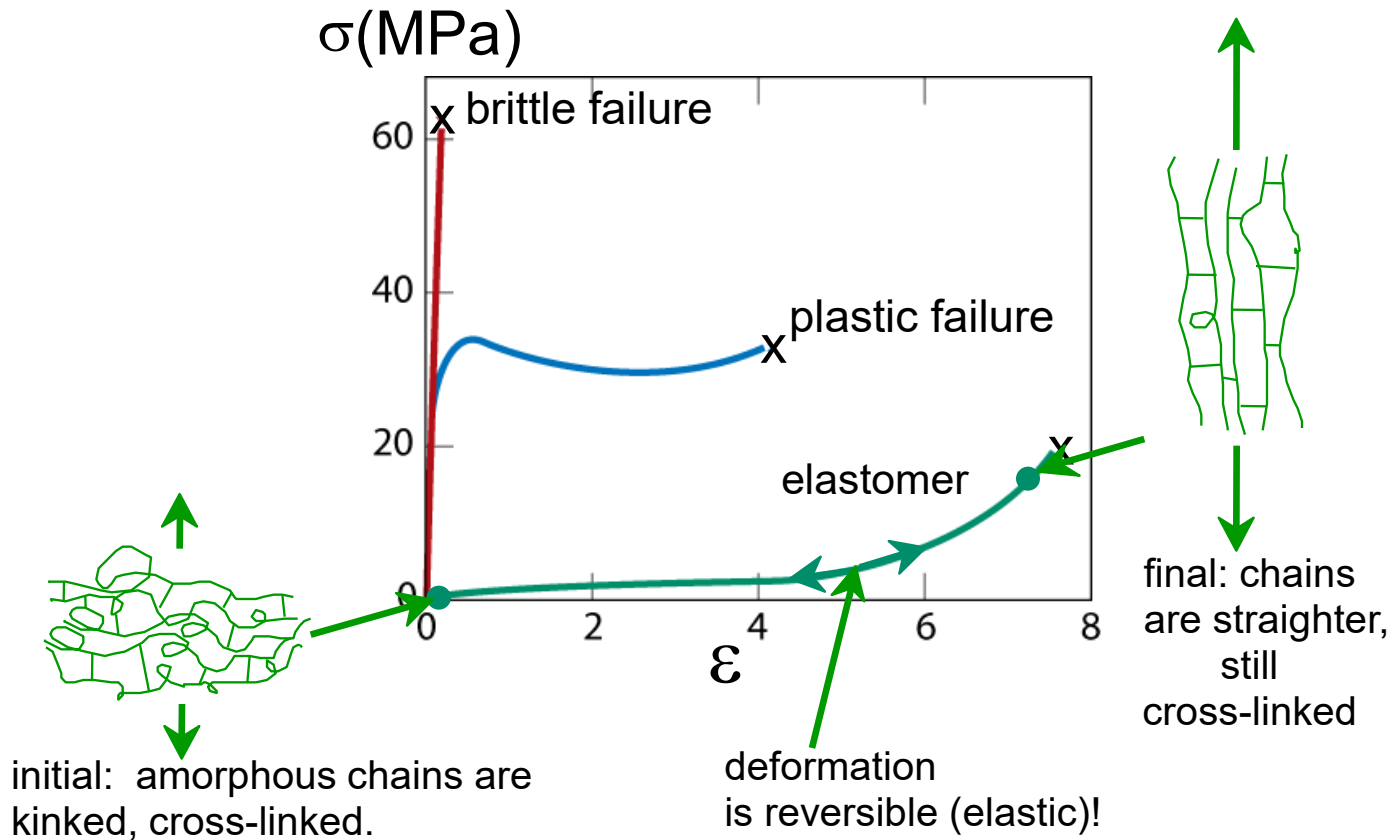


-- plastic behavior (of semicrystalline polymers)

Tensile test of a plastic polymer



Mechanisms of Deformation—Elastomers

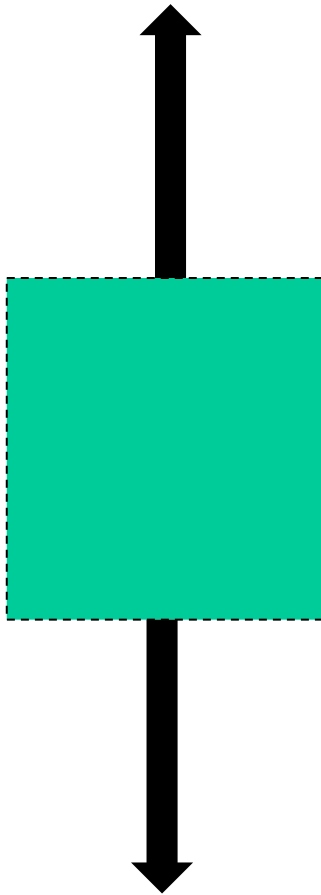


Rubbers are a subset of elastomers.

Elastomers = macromolecular material which returns rapidly to approximately its initial dimensions and shape after substantial deformation by a weak stress and release of the stress

What about Compression Loading?

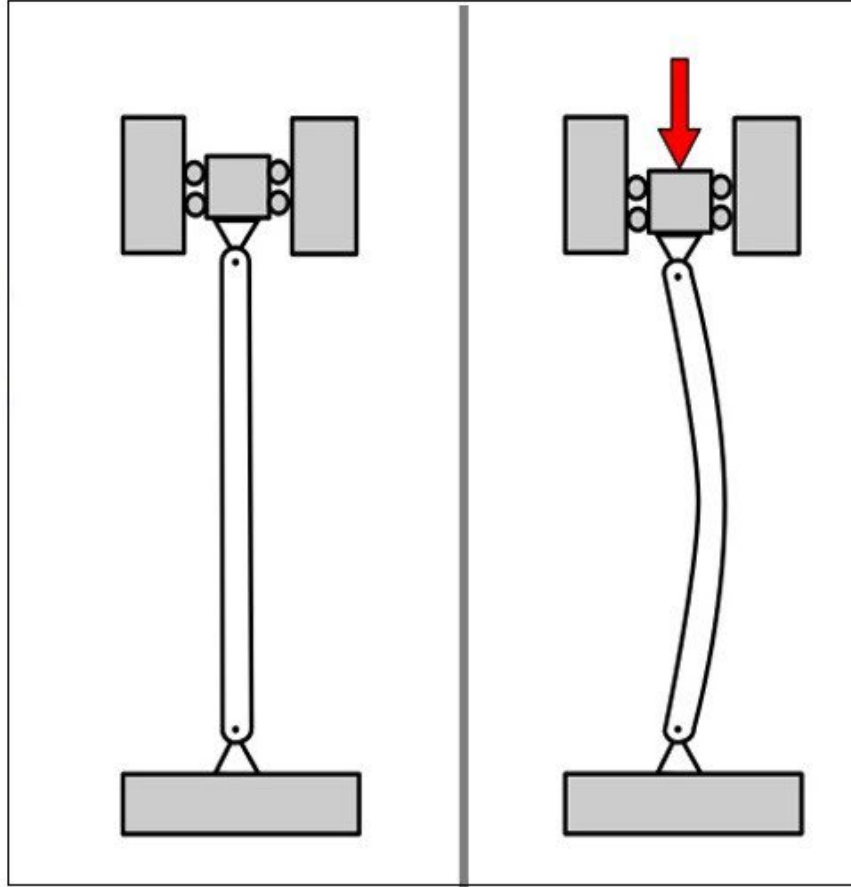
Tension



Compression

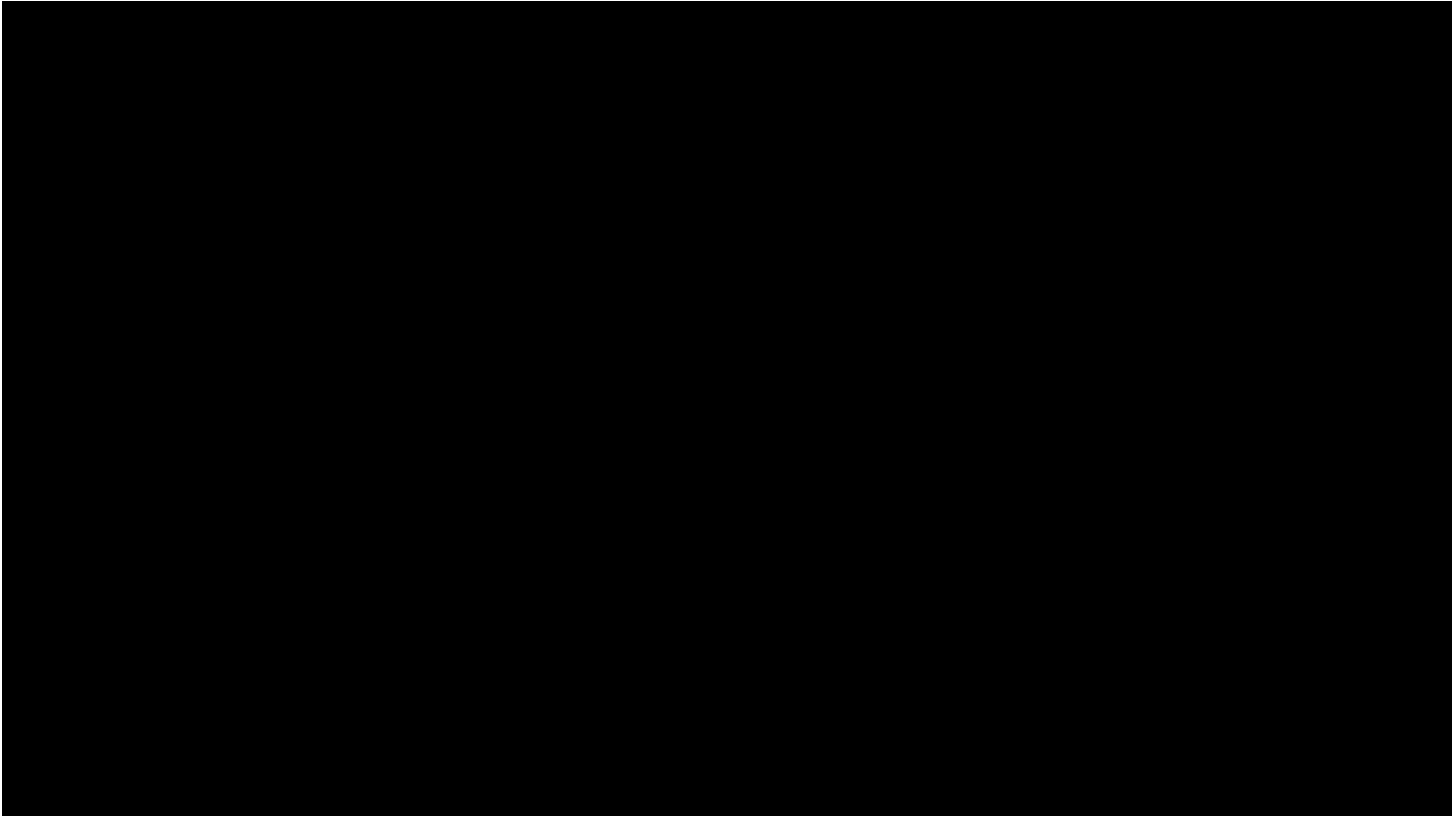


Ductile Materials Response to Compression Loading: Buckling



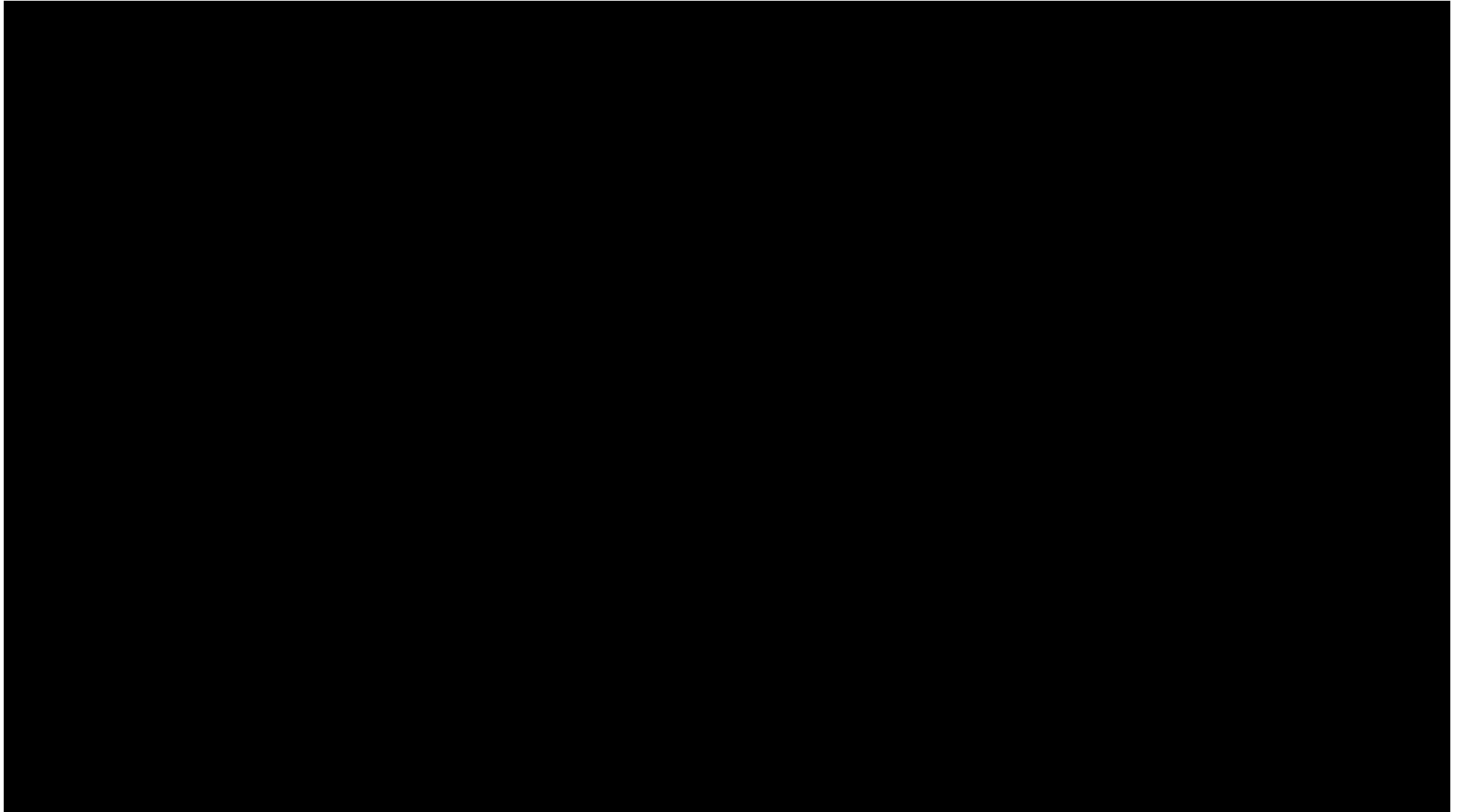
A high-aspect ratio cylindrical specimen: **buckle**

Low Slenderness



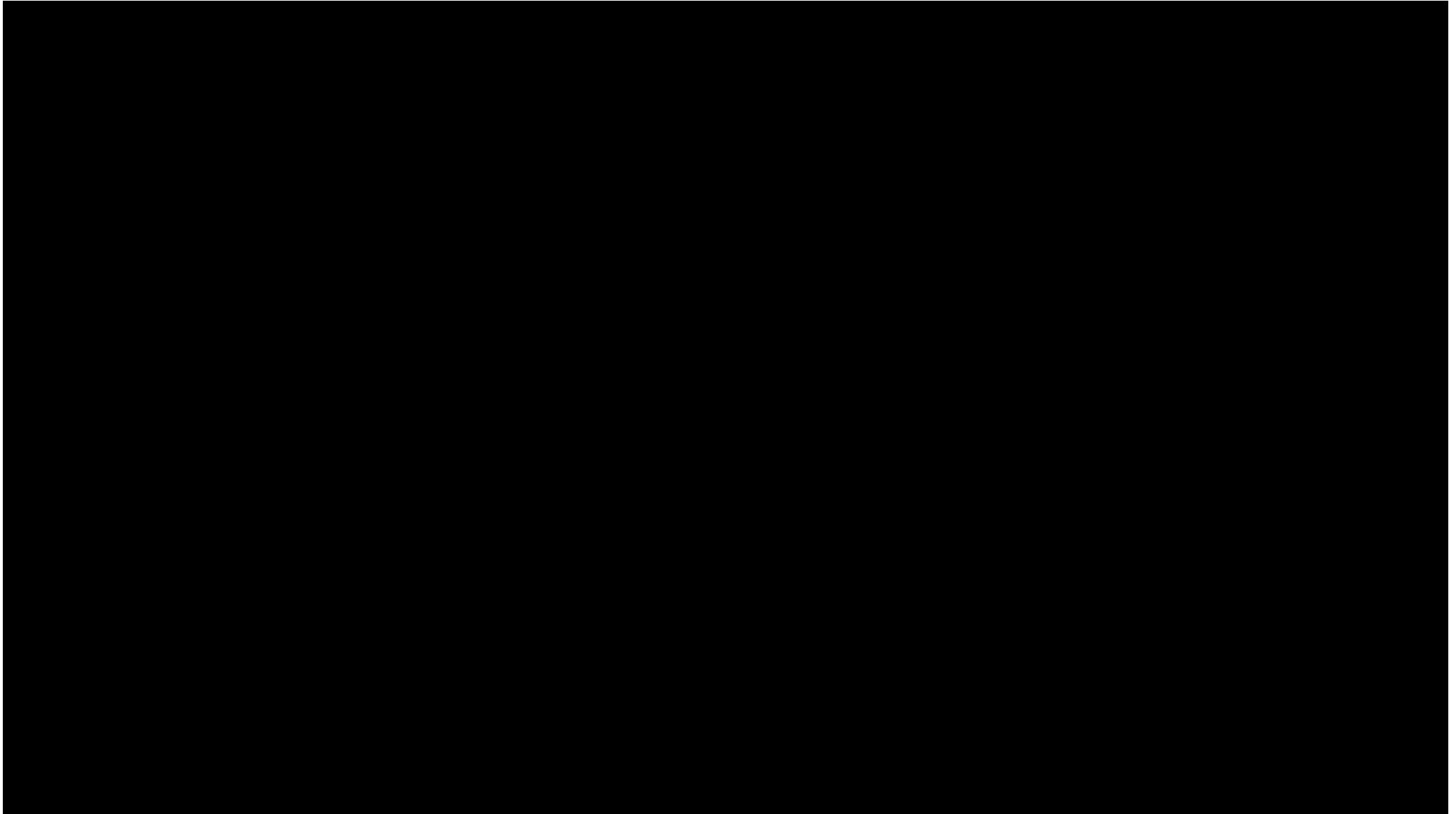
<https://youtu.be/jNwwub87l8o>

Intermediate Slenderness



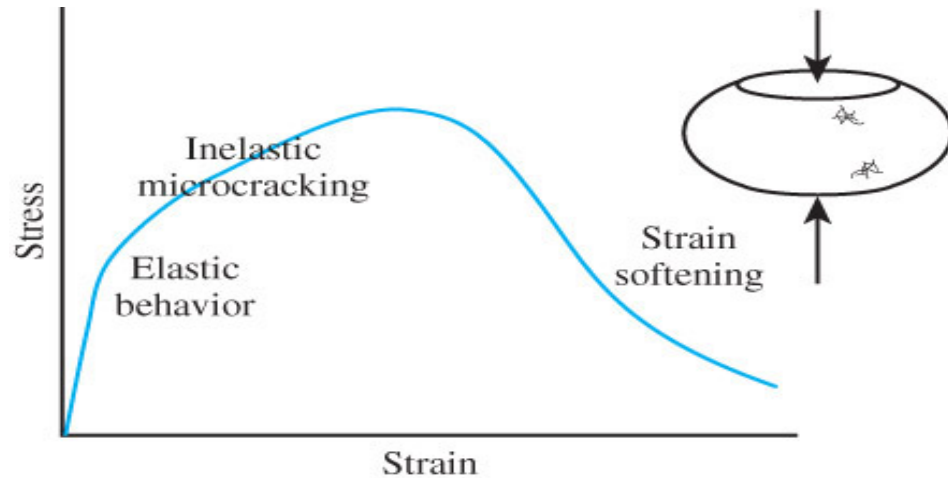
<https://www.youtube.com/watch?v=xCMUOvwyn8w>

Slender Steel Loading



<https://youtu.be/Mm7gYl18Muk>

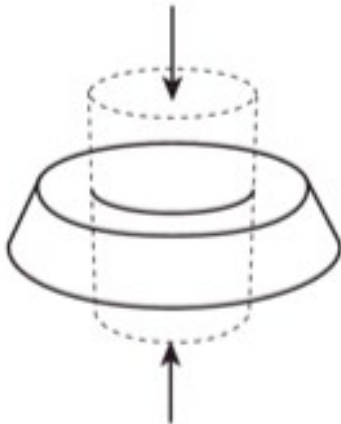
Ductile Materials Response to Compression Loading: Barreling and Pancaking



A low-aspect ratio cylindrical specimen:

- Barreling or pancaking

Due to Poisson's effect and ends constraint.

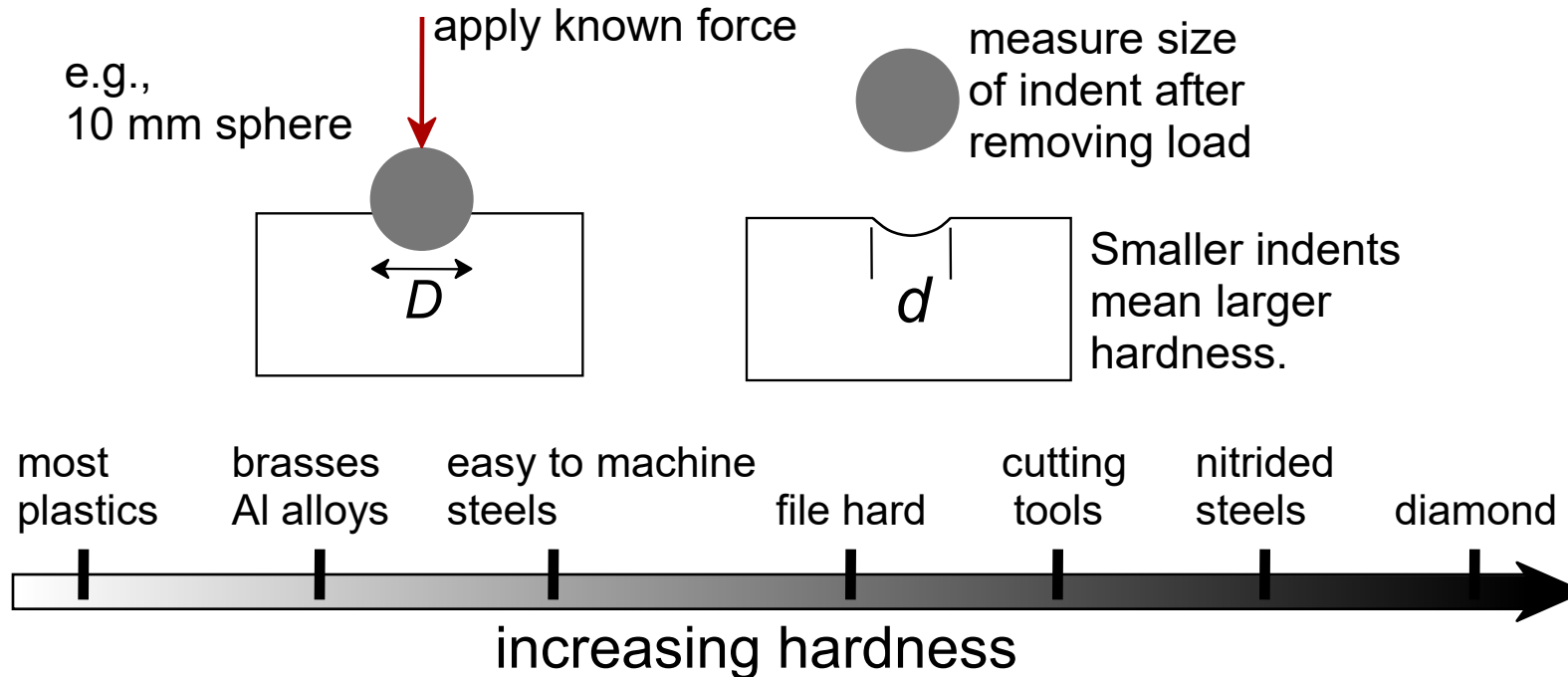


Brittle Materials Response to Compression Loading: Cracking

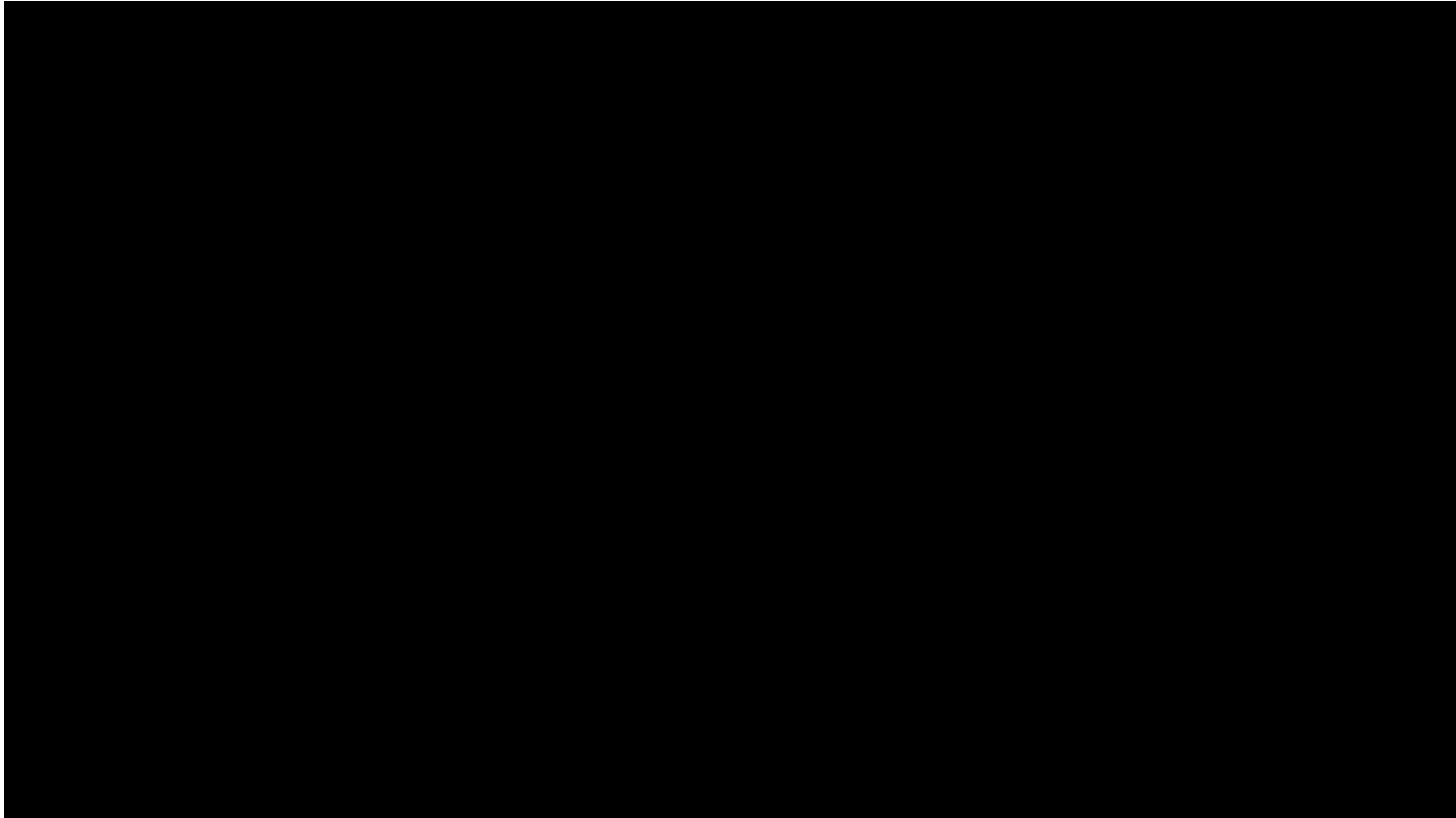


Hardness

- The resistance a material to scratching, indentation, or penetration
- Hardness is very important for manufacturing → Machining
- Hardness is a complex combination of strength, ductility, and stiffness.



Brinell Hardness Testing



<https://youtu.be/RJXJpeH78iU>

Table 7.2 List and Details of the Major Types of Hardness Tests

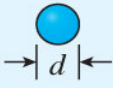
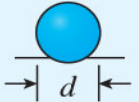
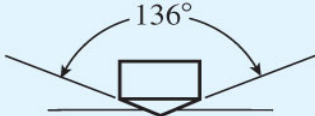
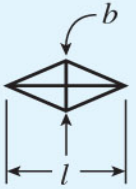
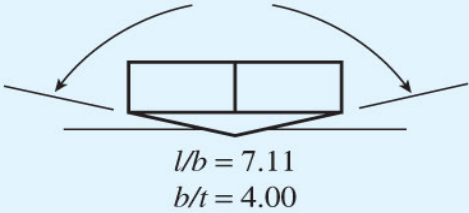
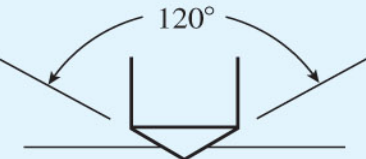
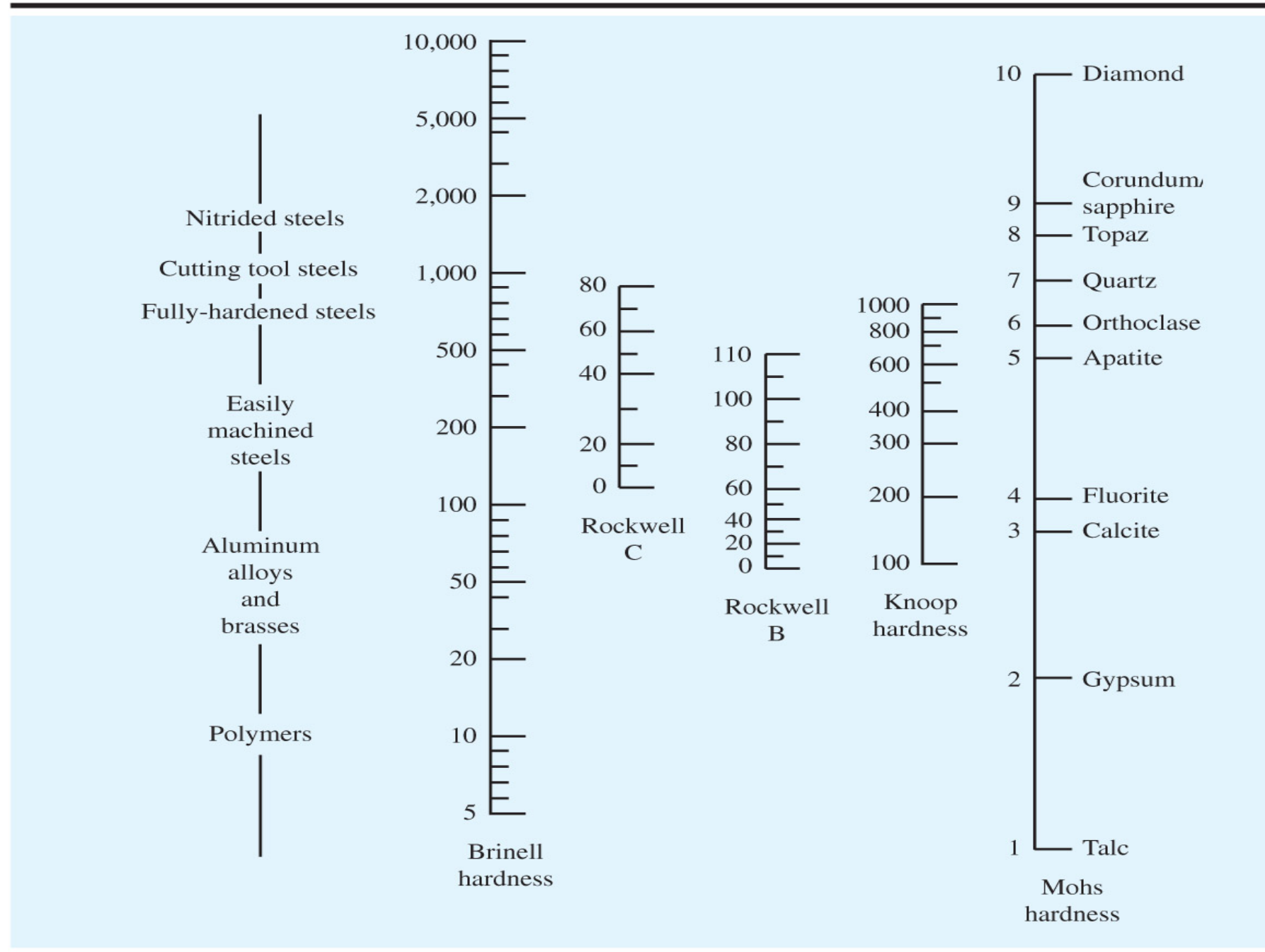
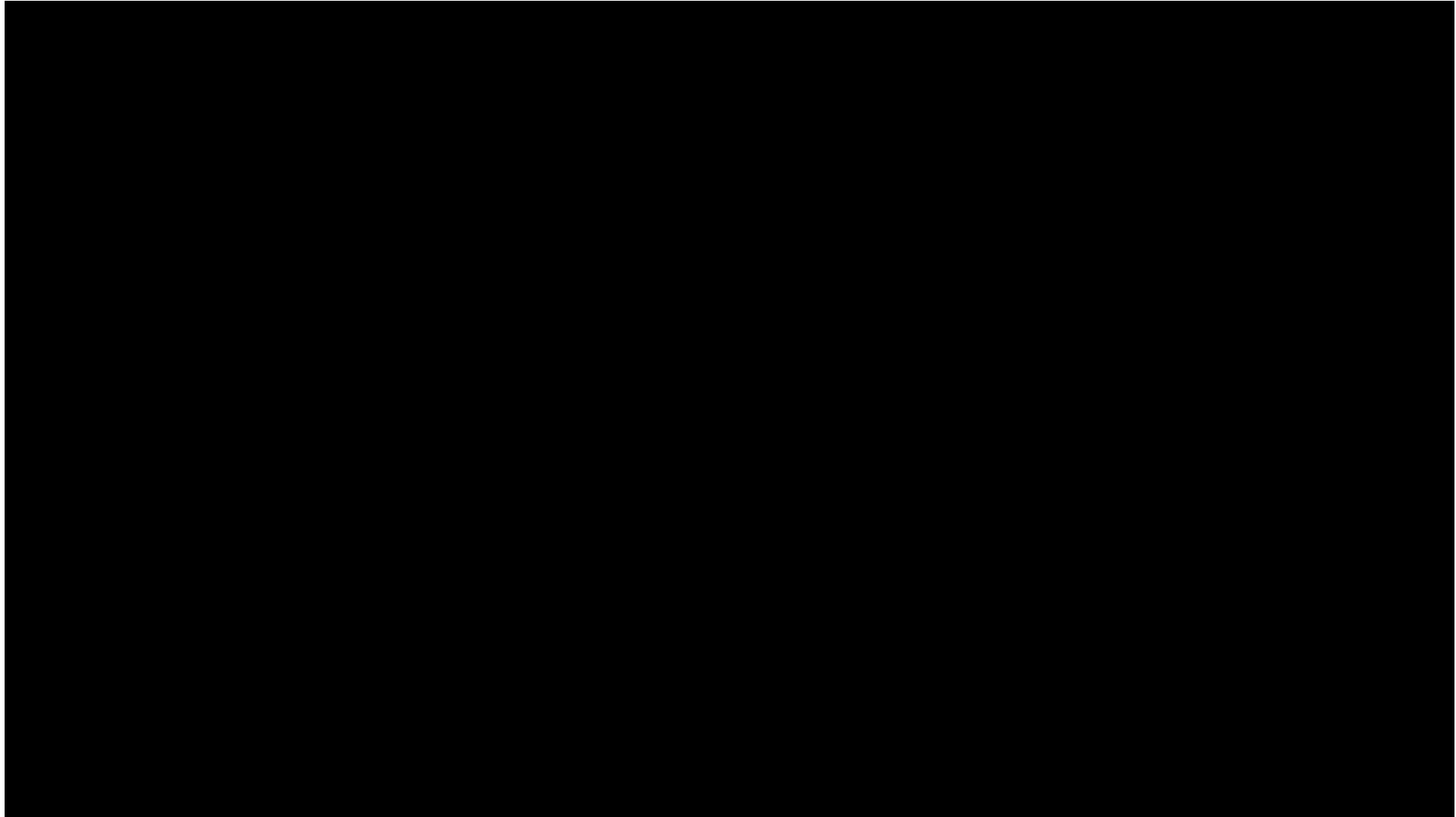
Test	Indenter	Indentation geometry		Load	Hardness number formula
		Top view	Side view		
Brinell	5- or 10-mm steel or tungsten carbide sphere			500–3000 kg	$HBN = \frac{2P}{\pi D(D - \sqrt{D^2 - d^2})}$
Vickers (micro)	Diamond pyramid			5–2000 g	$HVN = \frac{1.854P}{d_1^2}$
Knoop (micro)	Diamond pyramid			5–2000 g	$HKN = \frac{14.2P}{l^2}$
Rockwell	Diamond cone or $\frac{1}{16}$ -, $\frac{1}{8}$ -, $\frac{1}{4}$ -, $\frac{1}{2}$ -in. diameter sphere			60 kg 100 kg 150 kg	
Superficial Rockwell	Conical diamond or small sphere			15 kg 30 kg 45 kg	
Light load	Conical diamond or small sphere			7, 5, or 3 kg	
Micro	Small truncated cone			100, 500 g	
Macro	5- or 10-mm steel ball			500–3000 kg	

Table 7.5 Comparison of Several Popular Hardness Scales



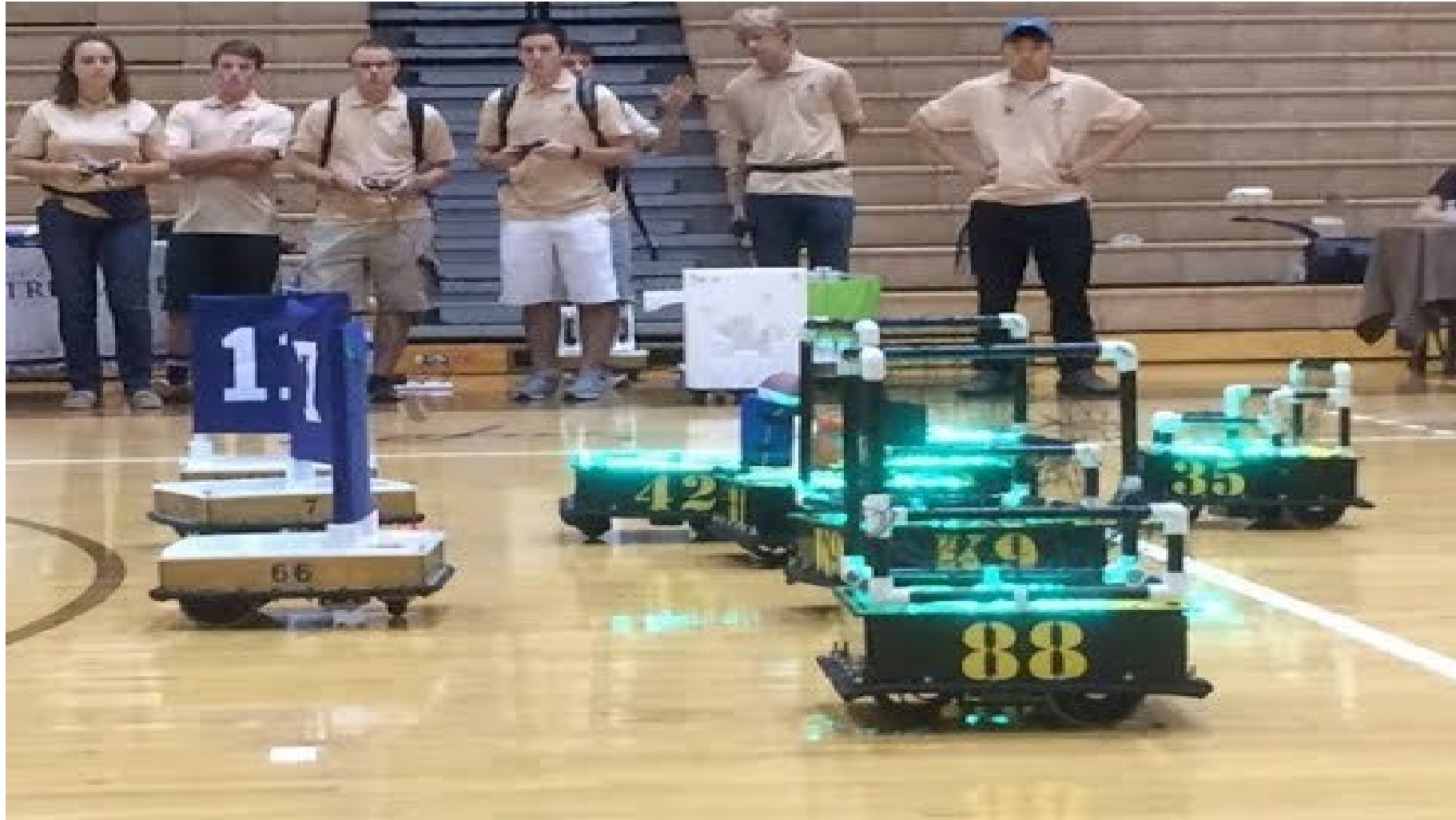
Mohs Hardness



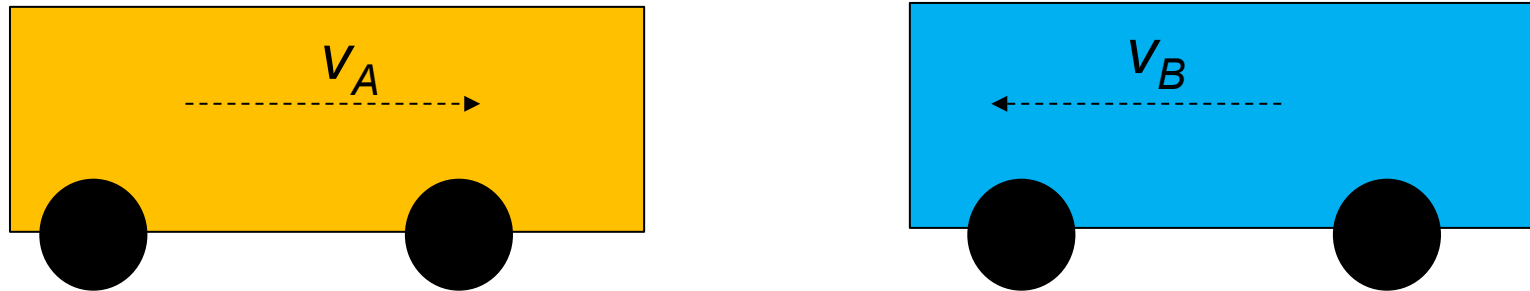
<https://www.youtube.com/watch?v=xZUgWzRpA-4>

Impact Testing

Robotic Football



Robot Collision:

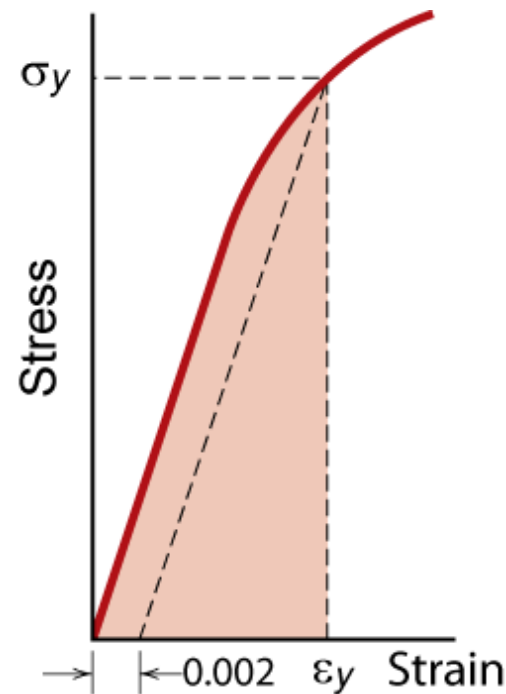


How can we know it won't break?

Impact

Resilience, U_r

- Ability of a material to store energy
 - Energy stored best in elastic region



$$U_r = \int_0^{\epsilon_y} \sigma d\epsilon$$

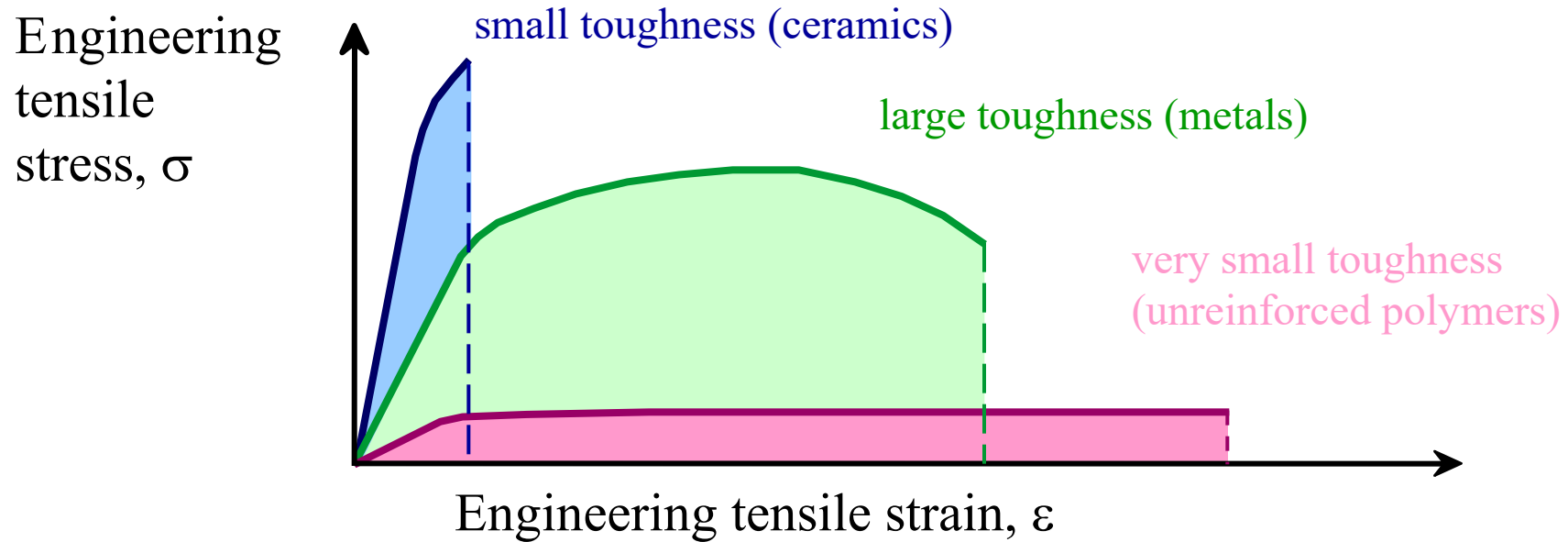
If we assume a linear stress-strain curve this simplifies to

$$U_r \cong \frac{1}{2} \sigma_y \epsilon_y$$

Adapted from Fig. 7.15,
Callister & Rethwisch 3e.

Toughness

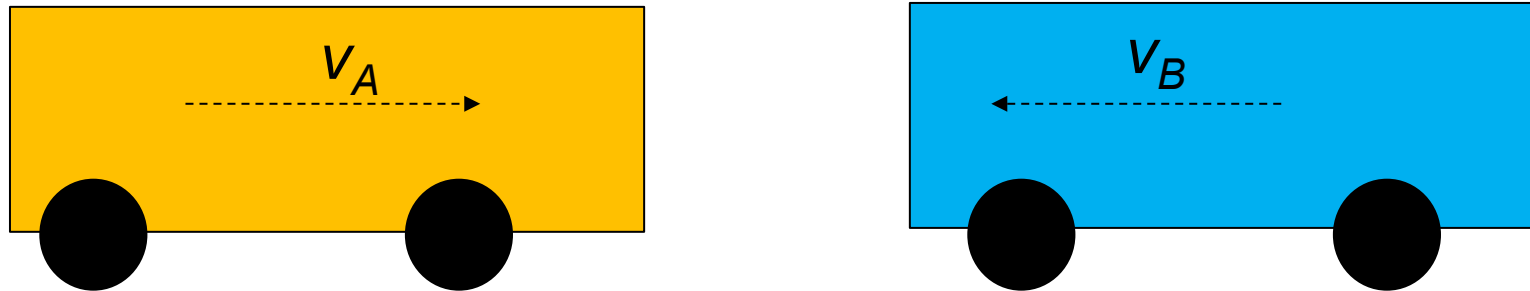
- Energy to break a unit volume of material
- Approximate by the area under the stress-strain curve.



Brittle fracture: elastic energy

Ductile fracture: elastic + plastic energy

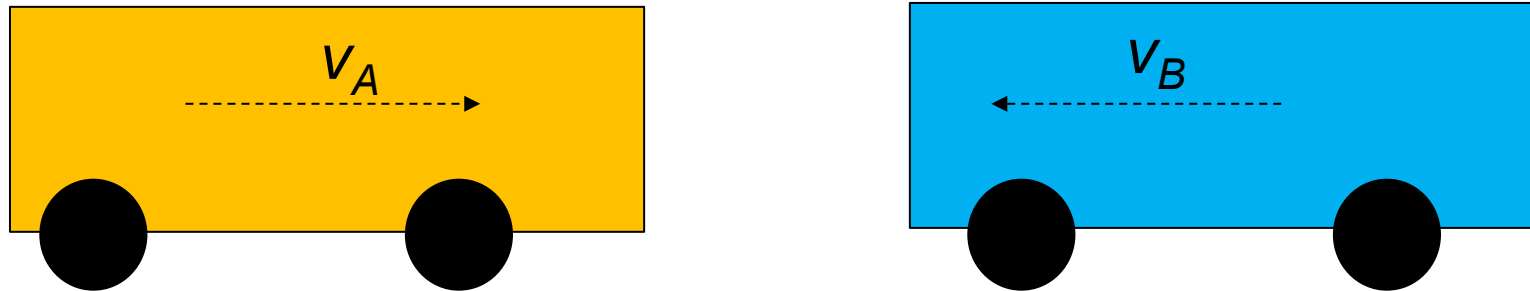
Robot Collision:



How can we know it won't break?

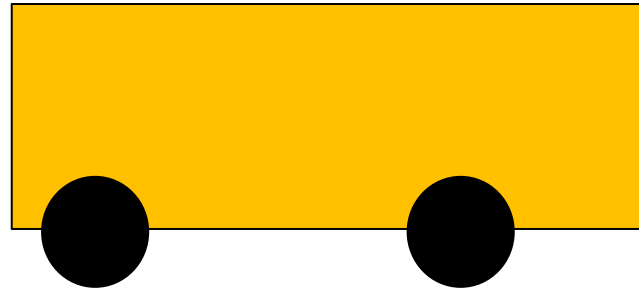
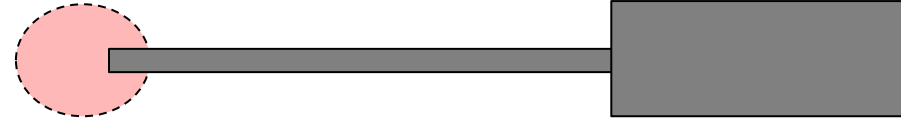
Robot Collision:

Moving robots have energy!



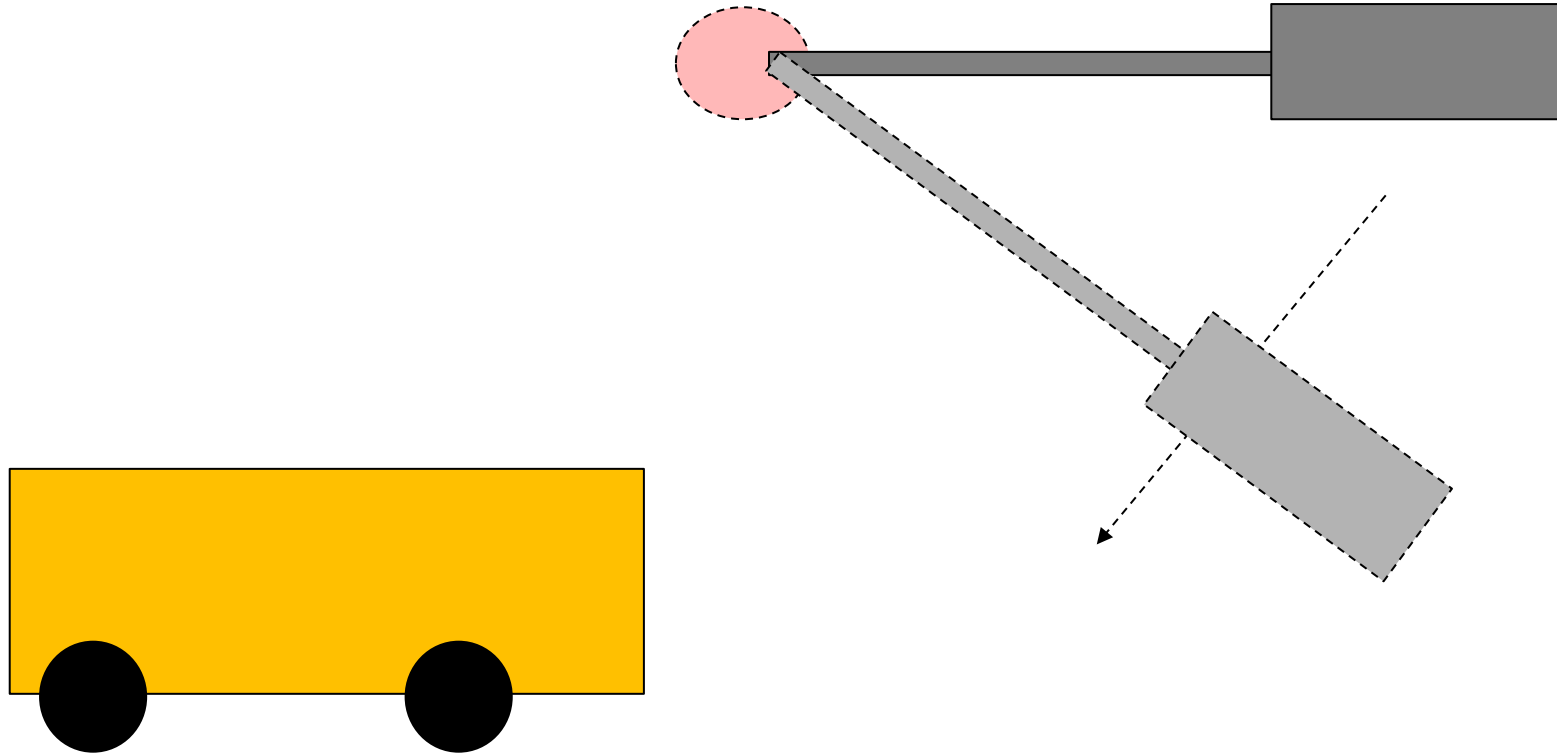
How can we know it won't break?

Robot Collision:



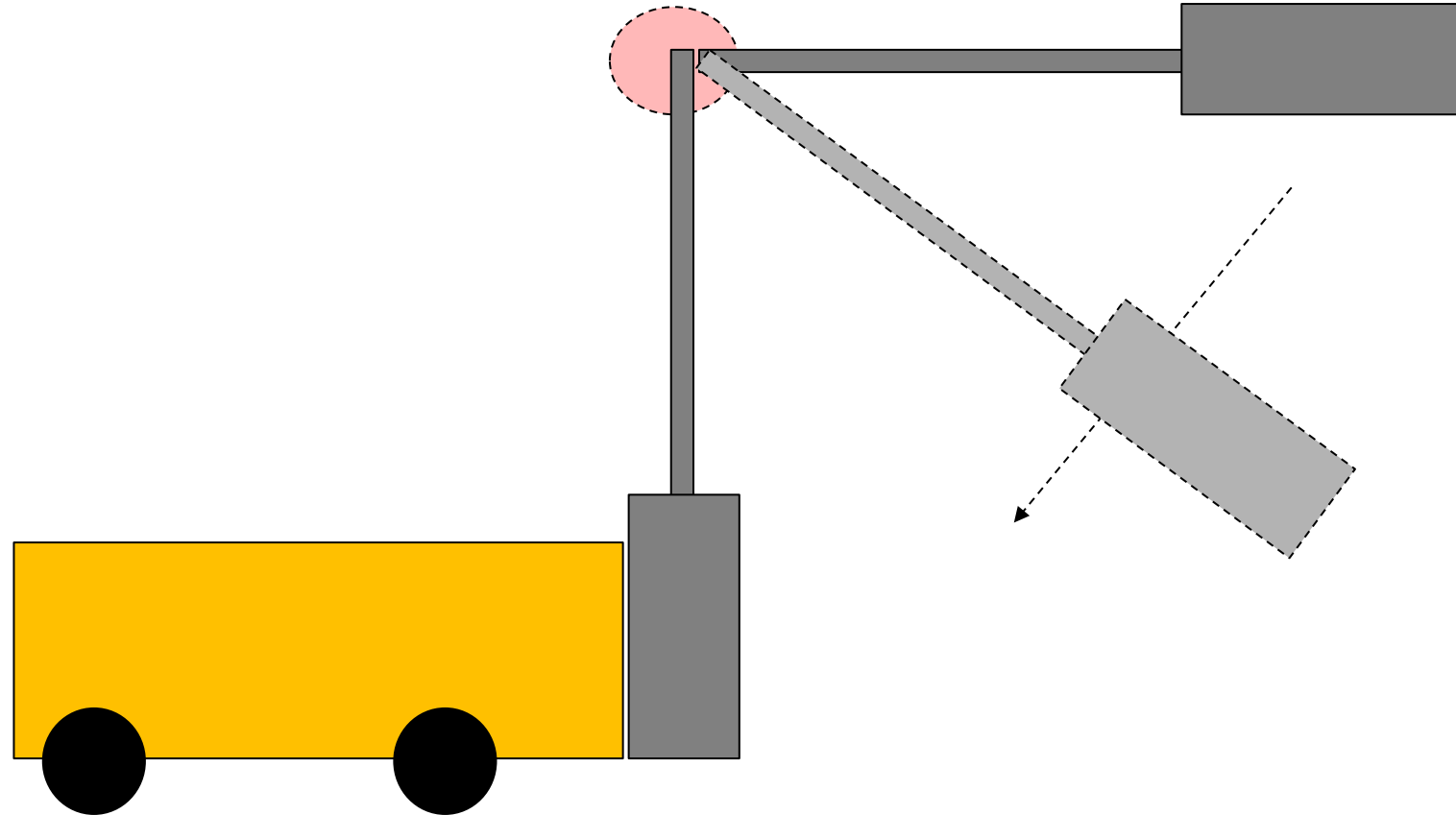
How can we know it won't break?

Robot Collision:



How can we know it won't break?

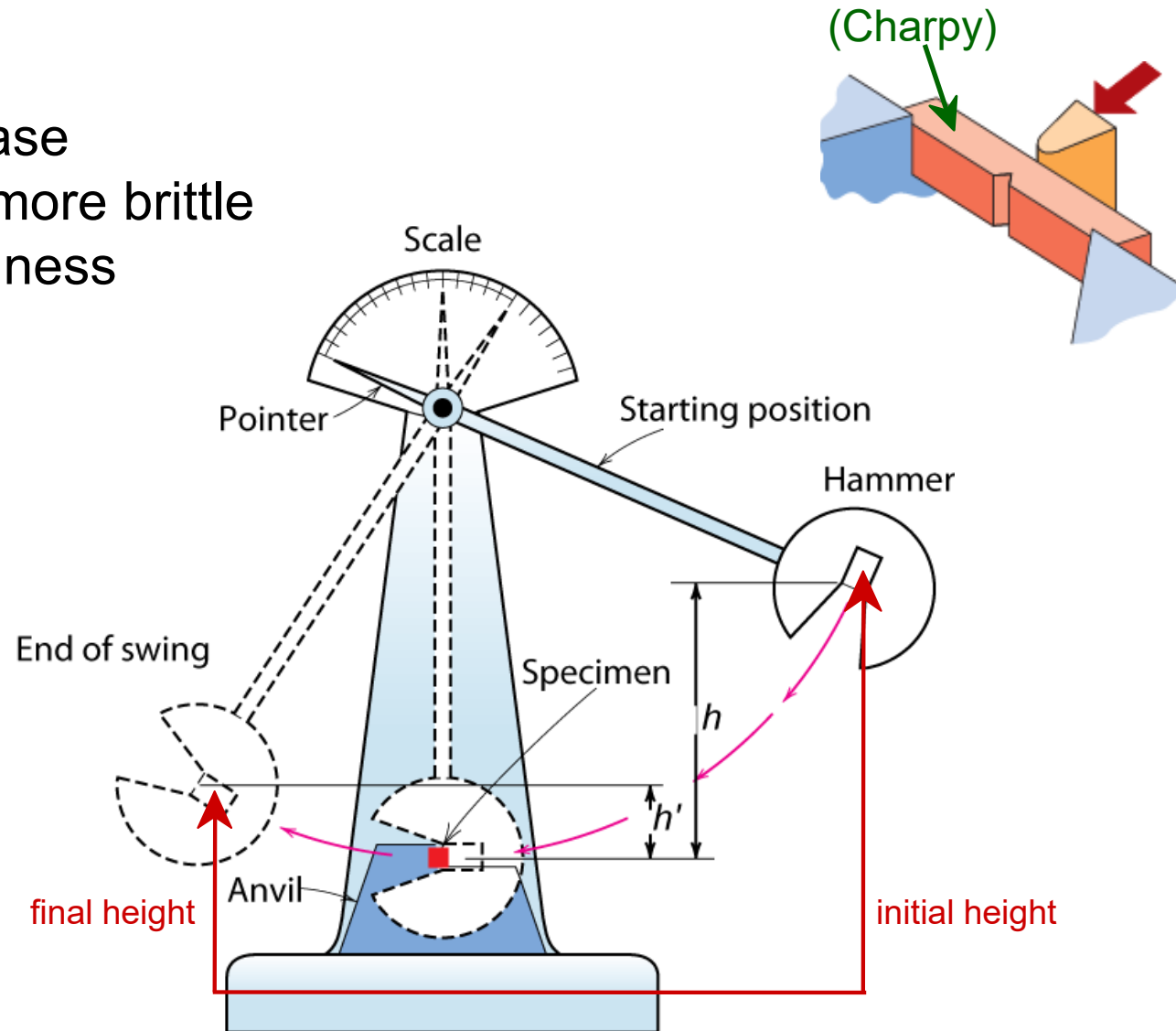
Robot Collision:



How can we know it won't break?

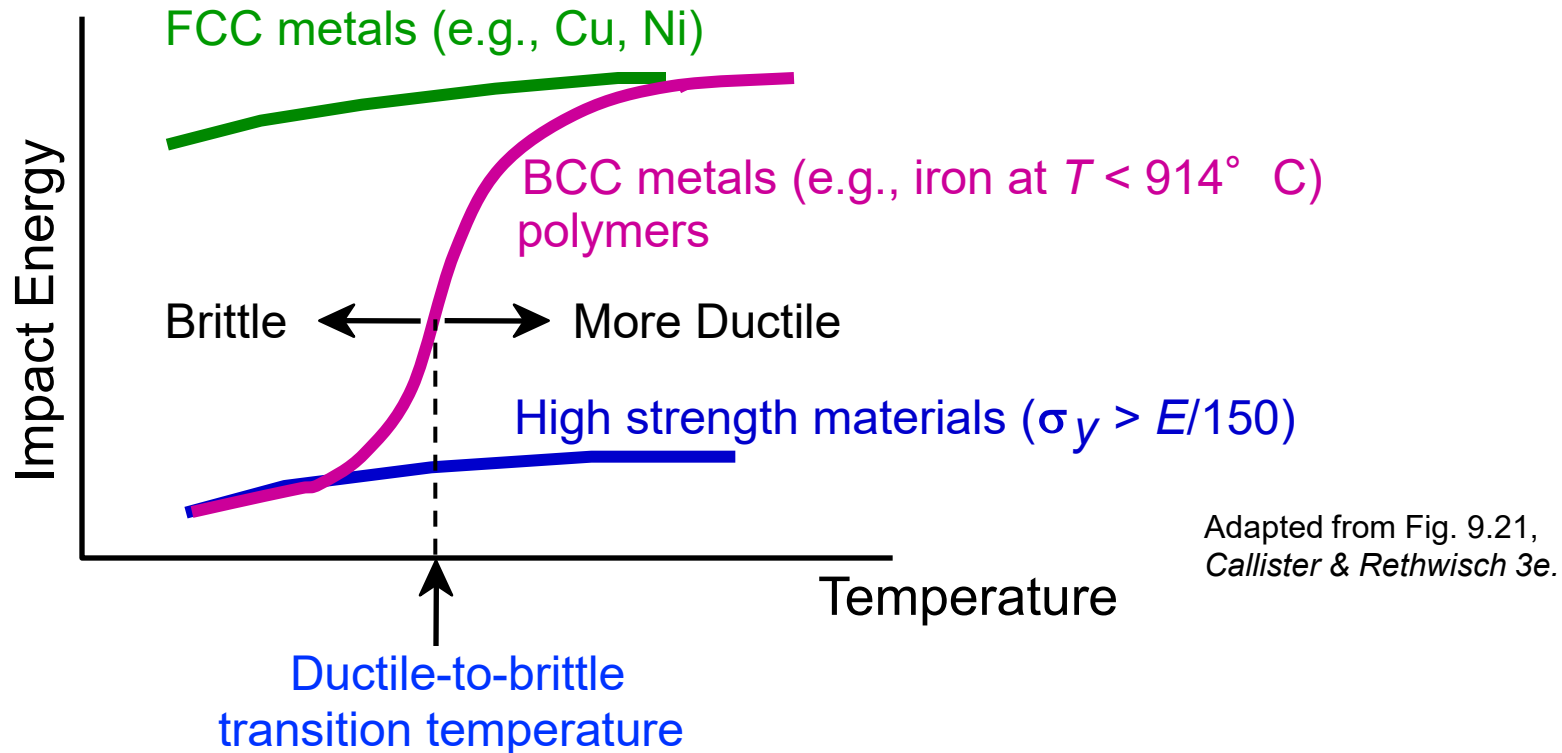
Impact Testing

- Impact loading:
 - severe testing case
 - makes material more brittle
 - decreases toughness



Temperature and Impact Energy

- **Increasing temperature...**
 - increases % EL and K_C
- **Ductile-to-Brittle Transition Temperature (DBTT)...**

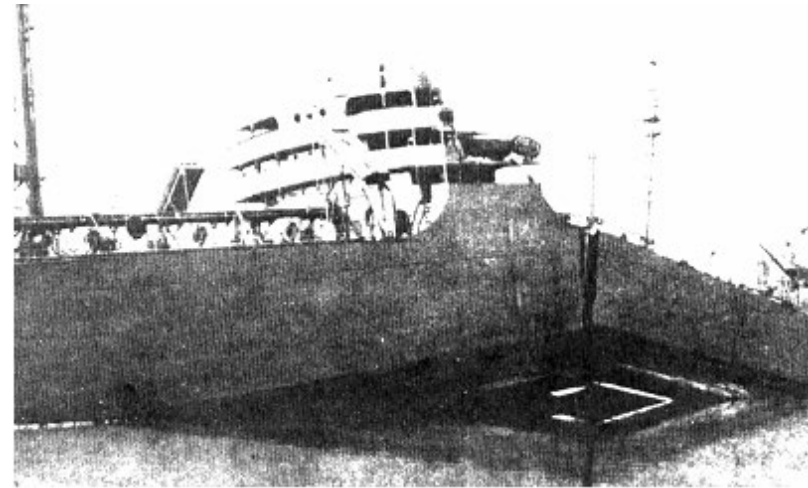


Design Strategy: Stay Above The DBTT!

- Pre-WWII: The Titanic



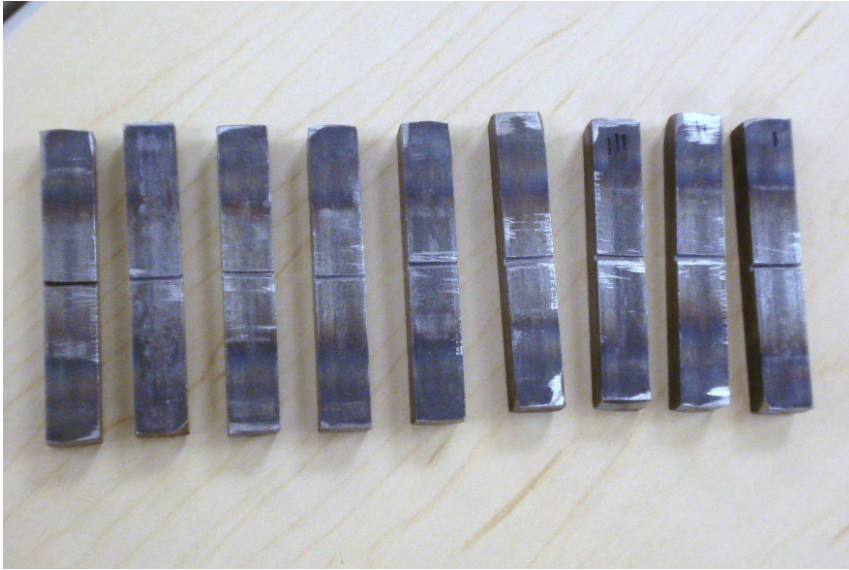
- WWII: Liberty ships



Problem: Used a type of steel with a DBTT $\approx$ Room temp.

DBTT?

For the following two materials, what would happen if you do impact testing on them at room temperature and liquid nitrogen temperature (-196°C)?



1018 Steel



Brass

(alloy of copper and zinc)